

Secret Code Lab



BIG IDEA Encryption scrambles a message so only someone with the secret key can read it. A longer, more secret key is a stronger lock.

1 HOOK 2'

2 ACTIVITY 8'

3 DISCUSS 3'

4 TAKEAWAY 2'

1 HOOK 2 MIN 60-second script “Mateo wants to pass a note to Amara across the room. But the note might get picked up on the way. So he writes it as a secret code: ‘SDXVH FKHFN WHOO.’ That looks like nonsense, right? But to Amara, who knows the secret key, it says something very important. (Pause.) When you visit a website with the little lock symbol, the message is scrambled so people on the way cannot read it. That scrambling is called encryption. Today you build a cipher wheel and become code makers and code breakers. This is a puzzle, and it is the very beginning of how real internet lockboxes work.”

2 ACTIVITY 8 MIN Build and Use a Cipher Wheel

- 1 (2 min) Get your wheel.** The outer ring is the real message; the inner ring is the scramble.
- 2 (1 min) Teach the key.** For Key 3, turn the inner ring until D sits under outer A. Find each message letter on the OUTER ring and write the INNER letter beside it: PAUSE becomes SDXVH.
- 3 (3 min) Decode.** “SDXVH FKHFN WHOO” (key 3), then “YJQQ F YWZXYJI LWTBSZU” (key 5).
- 4 (2 min) Make your own.** Encode a safety message with a key of your choice. Give your partner the message but NOT the key.

3 DISCUSS 3 MIN

- 1** “What is a key, and why must it be secret?”
- 2** “If a message is encrypted on its way, is the person you are talking to safe?” (No. It does not tell you who is on the other end.)
- 3** “Where else do we use secrets or keys?” (House keys, passcodes, passphrases.)

PREP IN 5 MINUTES

Materials: Cipher Wheel (R-4.1) or Cipher Strip, Code Messages (R-4.3), pencils.

- ☐ Pre-cut and pin one wheel (or strip) per pair.
- ☐ Copy the Code Messages.
- ☐ Know the answers: PAUSE CHECK TELL (key 3); TELL A TRUSTED GROWNUP (key 5).

4 TAKEAWAY 2 MIN “When someone asks for your key or secret, your hackles go up. A false alarm is fine. A code is only as strong as its key, and a key is only safe if it stays secret. Pause before you share. Check who is asking. Tell a grown-up.”

Hand signal: Hackles Up (hand brushed up the neck), Pause (palm), Check (magnifier fingers), Tell (hand on chest, then out).

EXIT TICKET “Decode ‘ULCLY ZOHYL FVBV WHZZDVYK’ (key 7), then write what a key is in your own words.” (NEVER SHARE YOUR PASSWORD.)

FUN BOOST Key Handshake (30 seconds). Partners invent a secret three-move handshake. That is their key. Can neighbors copy it without being taught?

LEVEL-UP Make a Master Key. Build your own scrambled alphabet and code a safety message for a younger class. Plan how to hand over the key safely.

DIFFERENTIATE Give a pre-written Key 3 chart and fewer letters per message; pair with a buddy.

FAMILY CARD Write each other a short coded note. Talk about what stays secret in your family and who may know it.

IF A STUDENT TELLS YOU SOMETHING

- **Say:** “Thank you for telling me. You did the right thing.”
- “Some secrets are fun. If a secret makes your tummy feel funny, you are always allowed to tell.”
- “I can’t keep this a secret, because I need to get you help. I will tell you who I tell.”
- **Do not say:** “I promise I won’t tell anyone.”
- “Tell me everything.”
- Anything beyond “Can you tell me more?”

First 10 Minutes: see the Disclosure Guide.

Then: tell your counselor or designated person right away. Follow your school policy and your state’s reporting law.